What Does Public-Data ESG Measure? A Transparent Multi-Factor Mid-Cap Index, Its In-Sample Validation Pitfalls, and an Out-of-Sample Test

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***Abstract*—Open research pipelines increasingly build environmental, social and governance (ESG) scores for mid-cap firms from public data. We ask what such scores measure and how they should be validated, using a reproducible nine-factor ESG–financial index for 269 US and Indian mid-caps in which every data cell records its source. Construct validity is weak: only 21 of 34 ESG indicators carry information, most E and S cells are financial proxies, and the other factors explain $R^2 = 0.42$ and 0.51 of those two pillars. Common in-sample evidence, namely trailing-return information coefficients (ICs), a contemporaneous "forward" proxy and a misapplied bootstrap, made the index look strong. In a hold-out window (2026-04-02 to 2026-09-23, $N = 260$) with hypotheses fixed in advance, the composite does not predict returns (IC $= -0.094$, 95% CI $[-0.219, 0.036]$), and the IC is negative under 96.7% of random weightings. Higher ESG does predict lower volatility beyond trailing volatility (partial $\rho = -0.163$, Holm $p = 0.013$), and the effect survives 8 risk controls and sector effects. The signal comes from the proxy-built S pillar, while measured ESG data show none of it (partial $\rho = 0.032$); it adds 0.0056 to out-of-fold R^2 and weakens given recent volatility. An ESG-only top-19 portfolio trailed the universe by 19.9 percentage points. Public-data ESG therefore behaves as a slow-moving low-risk tilt. We pre-register a confirmatory second window, and a single command regenerates every number in the paper.**

***Index Terms*—ESG measurement, construct validity, composite indicators, out-of-sample validation, pre-registration, mid-cap equities, reproducibility**

I. Introduction

Sustainable-investment assets reached about $30 trillion in the markets covered by the most recent global review [1], yet the environmental, social and governance (ESG) ratings that steer this capital disagree strongly across providers, with an average pairwise correlation of about 0.54 [2], [3]. The ratings are also proprietary, so their construction cannot be inspected [4], [5]. The problem is most acute for mid-caps. Large caps increasingly face mandatory sustainability reporting, whereas mid-caps are followed by fewer analysts and data vendors even though they are large enough to be investable.

A natural response, now common in open-source quantitative research, is to build ESG scores from public data. Such pipelines combine regulatory filings, free governance scores and financial-statement ratios, each justified by an economic rationale such as "higher operating margin within a sector implies lower emissions intensity" [6]–[8]. They are transparent, but transparency does not establish validity. Two questions are rarely asked of them: what the resulting score measures once each input is traced to its actual source, and whether the evidence usually offered for such indices can distinguish a good composite from a bad one.

We study both questions on a fully reproducible system: a nine-factor ESG–financial index for 269 mid-cap equities in the US and India, scored on a data snapshot taken at the close of 2026-04-02. The index weights its ESG pillars by Sustainability Accounting Standards Board (SASB) materiality [9], [10], combines them with eight financial factors under three investor profiles, and follows composite-indicator practice [11]. Every data cell carries a source label, and every number in the paper is generated from the pipeline outputs. An end-to-end audit of the pipeline found implementation errors that changed the scores materially, most importantly two indicators whose direction was flipped twice, and a family of validation designs that report favourable numbers whether or not the index works. We corrected the former and replaced the latter with an out-of-sample (OOS) test whose universe, weights and hypotheses were fixed before any post-snapshot price was downloaded.

The study answers four research questions. RQ1 asks what a public-data ESG score for mid-caps measures, and how much of it is financial information already contained in the other factors. RQ2 asks how far common in-sample validation designs overstate the quality of such a composite. RQ3 asks whether, over the following 118 trading days, the composite predicts returns and its ESG component predicts risk. RQ4 asks whether any risk signal reflects ESG information, a known risk characteristic, or past volatility carried forward. Figure 1 maps the study design onto these questions.

The paper makes four contributions.

1) *A construct-validity audit of public-data ESG.* Only 21 of 34 declared indicators carry information, and the E and S pillars are largely re-expressed financial ratios. Governance, built from Institutional Shareholder Services (ISS) scores and ownership data, is the only pillar

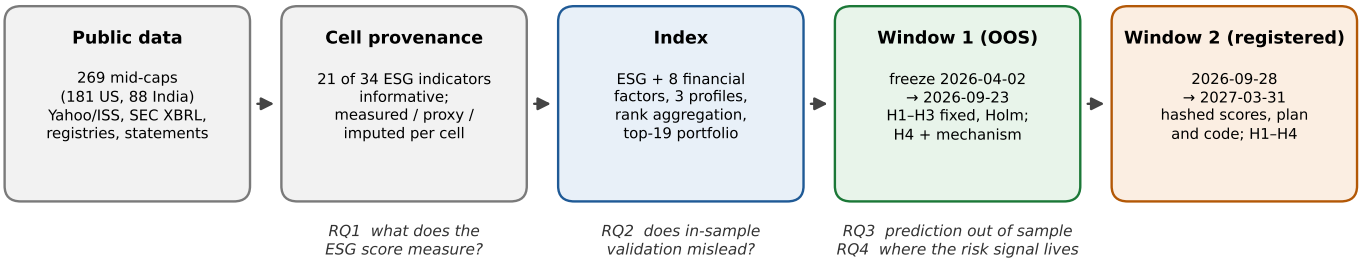


Fig. 1. Study design. Scores are frozen at the 2026-04-02 close and evaluated on the following window (RQ3–RQ4). A second window, whose scores, plan and code were hashed before it opened, provides a confirmatory test.

close to orthogonal to the financial factors (Section V-A).

2) *A quantification of in-sample validation pitfalls.* Trailing returns, a contemporaneous "forward" proxy, a correction fitted to the validation target and a misapplied bootstrap made a composite with no OOS predictive power look strong (Section V-B).
3) *A hold-out test with hypotheses fixed in advance.* The composite does not predict returns under the deployed or almost any other weights, whereas its ESG component predicts lower volatility. Tracing that signal to its data sources shows that it comes from financial proxies and recent risk, not from measured ESG data (Sections V-C to V-E).
4) *An open, provenance-tracked pipeline with a pre-registered second window.* The scores, analysis plan and code for the next six months were hashed and published before that window opened, which makes it a confirmatory test (Section IV-D).

The results are negative for returns and qualified for risk. The risk finding agrees with the view that ESG operates mainly through risk [12], [13], but in our data the part of the score responsible is financial information and slow-moving volatility. The results also bear out warnings about false discoveries and backtest overfitting in factor research [14], [15]: a transparent composite can still be invalid, and only out-of-sample evidence, traced back to its data sources, separates the two.

II. Related Work

A. ESG Rating Divergence and Measurement

Commercial ESG ratings disagree substantially. Berg, Kölbel and Rigobon [2] attribute the divergence mainly to measurement and scope differences rather than weighting. Chatterji et al. [3] document low convergence, and disagreement itself carries return and risk implications [16], [17]. Christensen, Serafeim and Sikochi [5] show that more disclosure can *increase* disagreement, and Kotsantonis and Serafeim [4] catalogue data problems such as inconsistent peer sets, imputation and scale. This literature treats ESG ratings as noisy, model-dependent measurements. We apply the same lens to a public-data ESG score, where every modelling choice is visible and can be audited cell by cell.

B. ESG, Returns and Risk

Meta-analytic evidence reports predominantly non-negative ESG–performance associations [18], [19]. Materiality matters: industry-material issues drive the effect [9]. Equilibrium models imply that investor tastes for sustainability can *lower* the expected returns of green assets while making them hedges [20], [21], and carbon exposure is priced [22]. On the risk side, corporate social responsibility (CSR) is associated with lower systematic risk [12], and high-trust firms fared better in the 2008–2009 crisis [13]. A risk association alone does not identify an ESG effect, however. Low-volatility and low-beta stocks [23], [24] and high-quality firms [25] have distinct risk and return profiles, and realised volatility is highly persistent. An ESG score that loads on these characteristics will "predict" risk without containing ESG information. We therefore test the risk channel against these characteristics and against recent volatility, and trace it back to the sources of the score's inputs.

C. Public-Data and Proxy ESG

Where direct disclosure is sparse, researchers infer ESG-relevant behaviour from observables. Examples include environmental performance and firm value [6], eco-efficiency [26], [27], employee satisfaction [7], entrenchment indices [28], machine-learned ESG scores [8] and text-mined sustainability reports [29]. Each proxy has a rationale, but aggregating many proxies built from the *same* financial statements risks producing an ESG score that is a relabelled financial-quality score. We quantify this risk directly (Section V-A).

D. Multi-Factor Models and Composite Indicators

Factor models explain the cross-section of returns with a small number of characteristics [30]–[34]. Commercial ESG indices tilt or exclude constituents on vendor ESG ratings [35], and the channels through which those ratings relate to valuation and risk have been studied directly [36]. Our index is a multi-criteria *ranking* rather than a pricing model. Its construction follows the OECD handbook for composite indicators [11]: normalisation, weighting, aggregation and sensitivity analysis.

E. Validation, Multiple Testing and Overfitting

Hundreds of published factors fail to replicate once multiple testing is taken into account [14]. Backtests that choose

among many configurations overfit with high probability [15]. The remedy proposed across empirical sciences is to fix the hypotheses and the analysis plan before outcomes are observed [37]. We adopt this remedy in two steps. The first is a hold-out window with hypotheses fixed before outcomes were computed and a family-wise error correction [38]. The second is a formally pre-registered window whose scores, plan and code were hashed and published before it opened. We also show concretely how in-sample designs common in index research produce favourable evidence irrespective of predictive validity.

III. Data and Index Construction

A. Universe and Snapshot

The candidate universe contains 321 firms: mid-cap constituents of the S&P MidCap 400, the Russell Midcap and the NIFTY Midcap 150 [39], plus 45 S&P 500 large caps that serve only as a reference group in normalisation. Fundamentals and trailing market statistics come from Yahoo Finance [40], [41] and SEC EDGAR [42] and are dated at the close of 2026-04-02, which we verified by matching every cached last price to the exchange close. A firm must have traded on the snapshot date to be scored. This filter removes 7 firms (ANSS, SAMSN, CMA, CHX, ZI, WELSPUNIND, SUVENPHAR) that had already been acquired or delisted and whose inclusion would have let the evaluation window "see" corporate events that were already known. The scored universe has 314 firms, of which 269 are mid-caps (181 US, 88 India) in 11 sectors; all analysis concerns the mid-caps. Indian monetary fields are converted at the 2026-04-02 close of INR/USD (92.97).

B. Public-Data ESG with Cell-Level Provenance

The ESG schema declares 34 indicators (11 E, 11 S, 12 G). Each company–indicator cell is filled by the first available source in a fixed order, and the source is recorded. The order runs from direct measurement to imputation: the EPA Toxics Release Inventory (TRI) for US facilities; ISS governance QualityScores redistributed by Yahoo Finance (audit, board, compensation, shareholder-rights and overall risk on a 1–10 scale, stored as $(10 - r) \times 10$ so that higher is better); SEC XBRL filing items; Yahoo ownership and analyst-coverage fields; membership of public initiatives (CDP, SBTi, RE100, UN Global Compact); within-sector percentiles of financial ratios, each with a published rationale; and finally the sector or global median. No random noise is added at any stage. Appendix A gives the source of each indicator.

Table I shows the outcome. Only 21 indicators vary across firms. No mid-cap matched an EPA facility record, and the Sustainalytics pillar scores that Yahoo once exposed are no longer populated, so the E and S pillars consist almost entirely of financial ratios and proxies. The G pillar draws mainly on ISS scores (39% of cells, available for 209 of 269 firms), followed by ownership and analyst-coverage fields.

TABLE I
Provenance of ESG cells used in scoring (% of cells per pillar, mid-cap universe). "Used/declared" counts indicators with cross-sectional variation.

Source of cell	E	S	G
Indicators used / declared	4/11	7/11	10/12
ISS governance score	–	–	39
Yahoo ownership/coverage field	–	–	28
SEC XBRL item	–	–	3
Public registry (0/1)	25	14	–
Financial ratio	–	14	–
Financial proxy	75	70	12
Imputed	<1	1	19

C. Factor Scores

Continuous variables are winsorised at the 1st/99th percentile (2.5th/97.5th for valuation ratios) and standardised with a robust z-score based on the median absolute deviation (MAD), $z = (x - \mathrm{med}(x))/(1.4826\,\mathrm{MAD}(x))$, clipped to $[-3, 3]$ [43]. For the operational, risk-adjusted, value, growth and stability factors, indicators are first converted to within-sector percentiles, and indicators that fail a Shapiro–Wilk normality test ($p < 0.05$) receive normal scores. The financial, operational, growth, value and stability scores are then blended 70:30 with their own within-sector percentiles. Lower-is-better indicators are sign-flipped only when they are stored as raw magnitudes; Section III-E shows why this matters.

Within the ESG block, indicators are averaged within categories (e.g. emissions, labour, board), categories are weighted into E, S and G, and the pillars are combined with sector-specific SASB materiality weights (Appendix B) [9], [10]. Governance values are quantile-harmonised by source before the G pillar is formed, so that ISS and SEC-derived inputs share a scale.

The eight financial factors are conventional. The financial factor covers profitability, margins, cash and payout; the operational factor revenue per employee, R&D intensity, free cash flow (FCF), gross margin and headcount; the risk-adjusted factor one-year Sharpe and Sortino ratios, maximum drawdown, Amihud illiquidity [44], skewness and kurtosis; the growth factor revenue, earnings and sustainable growth; the value factor inverted P/E, P/B, P/S, EV/EBITDA and EV/revenue; and the stability factor liquidity, leverage and 30-day volatility. A sector-position factor measures size and efficiency relative to sector peers. A market factor (liquidity, beta, volatility and 1- to 6-month momentum) is computed but receives zero weight in every profile, so that firms are not scored on the trailing returns that earlier versions of the pipeline used as the validation target.

D. Aggregation and Investor Profiles

For profile p with weights $w^{(p)}$, the preference score is a weighted mean of factor percentiles,

$$C_i^{(p)} = \sum_k w_k^{(p)} \, \frac{100 \ \mathrm{rank}(F_{ik})}{N}, \qquad \sum_k w_k^{(p)} = 1, \quad (1)$$

TABLE II
EFFECTIVE PROFILE WEIGHTS (%) AFTER REMOVING THE MARKET FACTOR.

Factor	ESG-first	Balanced	Financial-first
ESG composite	56.1	23.2	5.6
Financial	8.2	18.9	27.8
Risk-adjusted	8.2	15.8	20.0
Growth	6.1	14.7	16.7
Stability	6.1	8.4	7.8
Sector position	7.1	8.4	5.6
Operational	5.1	6.3	8.9
Value	3.1	4.2	7.8
Market	0.0	0.0	0.0

with average ranks for ties and median fill for missing factors. Ranks are taken over the $N = 314$ scored firms, so the 45 large caps act as a fixed reference group. Table II lists the weights of the three profiles; the balanced profile puts 23.2% on ESG. The index portfolio is the equal-weighted set of the top $\lceil 7\% \cdot N_{\text{mid}} \rceil = 19$ mid-caps ($N_{\text{mid}} = 269$).

The profile weights were set by the authors from in-sample rank correlations with *trailing* momentum, a practice we criticise in Section V-B, and were last revised in July 2026, after the snapshot. The OOS analysis therefore also reports the February 2026 weights, which predate the snapshot, and equal weights.

E. Audit Corrections

Before any OOS number was computed, we audited the pipeline and corrected its errors (Appendix C gives the full list). The most consequential was a double direction flip: the Scope 1 emissions and ESG risk-rating indicators were already stored as "higher is better" scores and were flipped again, which rewarded high-emission and high-governance-risk firms. The Spearman correlation between the pre- and post-audit E pillars is -0.45, and for G it is 0.38. The audit also removed multiplicative Gaussian noise from median imputation, a "monotonicity correction" that fitted each factor by isotonic regression to the proxy later used to validate it, an undocumented gate that set the bottom 30% of firms by financial score to zero, an exponent of 1.5 on preference scores, and a live, non-snapshot FX rate. After these corrections the balanced composite has a rank correlation of 0.95 with the pre-audit version, but only 13 of the 19 top names are shared.

IV. VALIDATION DESIGN

A. In-Sample Diagnostics

On the snapshot cross-section we report conventional diagnostics: variance-inflation factors (VIFs), principal component analysis (PCA) and agreement between profiles. We also re-run the pipeline's former validation designs to measure how much they inflate apparent quality. These are rank information coefficients (ICs) against 1-, 3- and 6-month *trailing* returns, which end on the snapshot date; ICs against a "forward quality proxy", which is in fact a *contemporaneous* composite of ROA, ROE, margins, growth and liquidity that shares inputs with several factors; and quintile monotonicity tests against the same proxy.

B. Out-of-Sample Test

Scores are fixed at the 2026-04-02 close, and outcomes are measured over (2026-04-02, 2026-09-23], which spans 118 US and 122 Indian trading days. From split- and dividend-adjusted closes we compute returns at one and three months and over the full window, together with realised volatility, maximum drawdown and beta against the S&P 500 or NIFTY 50. Returns are demeaned within country, because the two markets had different index returns (17.1% and 3.2%); sector-and-country neutralisation is a robustness check. 9 firms were acquired during the window (CFLT, CWAN, EXAS, HOLX, WBS, HLX, CTRA, SEE, EA) and have no end price, so we report bounds that assign them the 5th or 95th percentile return.

Three hypotheses were fixed before outcomes were computed. All are one-sided, with Holm correction across the family [38].

H1 The balanced composite has a positive rank IC with the country-neutral full-window return.

H2 Given trailing (2-year) volatility, higher ESG predicts lower realised OOS volatility (partial Spearman).

H3 The top-19 balanced portfolio beats the equal-weighted universe.

Confidence intervals for H1–H3 come from 5,000 firm-level bootstrap resamples (1,000–2,000 for secondary analyses), and in H3 the portfolio is re-formed in each resample. Secondary analyses cover factor ICs by horizon (Holm-corrected within horizon), portfolios by profile, and ESG–risk partial correlations and rank regressions with sector and country fixed effects and heteroskedasticity-robust (HC3) errors.

With $N = 260$, a two-sided test at $\alpha = 0.05$ has 80% power only for $|\rho| \geq 0.173$. ICs of the size usually reported for single factors (0.02–0.10 [45]) are therefore not detectable in one cross-section, and a null for H1 means "no large effect" rather than "no effect".

The window had already ended when the hypotheses were written down, and the design was not externally pre-registered [37]. Its safeguards are procedural: universe, weights and scores are frozen artefacts dated at the snapshot, post-snapshot prices were downloaded only after the hypotheses and code were fixed, and all alternative weightings are reported.

C. Secondary Analyses: Is the Risk Signal ESG?

If H2 holds, three alternative explanations must be ruled out before the effect can be attributed to ESG. These tests are exploratory and were specified after H1–H3 had been evaluated.

The first is that ESG proxies for a known low-risk characteristic [23]–[25]. We regress the percentile rank of realised volatility on the ESG rank and on the ranks of trailing volatility, trailing beta, size, quality (profitability score), leverage (debt/equity), liquidity (log dollar volume), six-month momentum and value (price/book), with sector and country

fixed effects and HC3 errors (*H4*). We also ask whether ESG improves an out-of-fold volatility forecast (10-fold cross-validation, 50 repeats), and we double-sort firms on trailing volatility (terciles within country) and ESG.

The second is that the signal comes from the financial proxies rather than from measured ESG data. For each firm we average the percentile ranks of its indicator cells separately by source: *measured* (ISS, SEC, public registry; 8.9 cells per firm on average), *proxy* (financial-statement proxies; 10.1) and *imputed* (sector or global medians; 1.9). We then re-run the risk tests on each sub-score.

The third is that ESG is a slow-moving record of past volatility. We split the window at 2026-06-28 and ask whether ESG predicts second-half volatility given *first-half* realised volatility, which the scores never saw but an investor would have.

Firms whose realised volatility fell below a quarter of their trailing volatility (PEN, AES) were trading at announced takeover prices. We report results without them, by country, with winsorised volatility, and without utilities and real estate. For the return null, we report the OOS IC over 2000 weight vectors drawn uniformly from the simplex and over a sweep of the ESG weight from 0 to 1, month-by-month ICs, and top-19 excess returns net of round-trip costs of 16 bps (US) and 50 bps (India, including securities transaction tax), and of three times those costs.

D. Pre-Registered Second Window

To make the next test confirmatory, we froze the 269 firm scores and a four-hypothesis analysis plan on 2026-09-25, before the second window (2026-09-28, 2027-03-31] opened. The plan, the frozen scores and the evaluation code are hashed (SHA-256), and the package was pushed to the public project repository on 2026-09-25; that push is its timestamp. At the time of writing the window is still open and no outcome in it has been computed. The plan keeps H1–H3, adds H4 (Section IV-C) and applies the Holm correction across all four. In H2 and H4 the controls are those observable at the anchor date, namely six months of realised volatility, beta and return. The evaluation code refuses to run before the window closes or if any registered file has changed. A dry run of the same code on (2026-07-01, 2026-09-23] is included in the registration.

E. Rank Uncertainty

Rank intervals combine two sources of uncertainty over $B = 1000$ draws. The reference universe used for percentiles is resampled with replacement, and the profile weights are drawn from a Dirichlet(κw) with $\kappa = 100$, about $\pm 20\%$ relative variation for a weight of 0.2. The re-implementation reproduces the deployed composite with Spearman $\rho = 0.999$.

V. Results

A. RQ1: What the ESG Score Measures

Little of the ESG schema is filled with ESG data (Table I). The E pillar has four informative indicators: within-sector percentiles of operating margin, revenue-to-market-value and gross margin, standing in for emissions, renewable share and waste, and one binary flag for public climate commitments. Six of the seven S indicators are financial; revenue per employee even enters twice, once directly and once as a proxy for employee satisfaction. The G pillar is different, because ISS governance risk scores fill 39% of its cells.

The consequence is that E and S largely duplicate the financial factors. Regressing each pillar on the other seven factor scores gives $R^2 = 0.42$ for E and 0.51 for S, against 0.03 for G, and the ESG composite has a Spearman correlation of 0.58 with the financial score. E and S are moderately correlated with each other ($\rho = 0.42$) and nearly unrelated to G ($\rho = 0.01$ and 0.12). The label "ESG" thus covers two different objects: a within-sector measure of financial quality (E and S) and largely independent third-party governance information (G). Conventional diagnostics miss this. The maximum VIF among factor scores is only 2.04 (ESG composite), and the first three principal components explain 64.9% of factor variance, which looks like a well-conditioned factor set; the redundancy sits inside the ESG block.

The three investor profiles also differ less than their names suggest. The balanced composite has Spearman correlations of 0.90 and 0.96 with the ESG-first and financial-first composites, and its top-19 list shares 14 and 14 names with them, even though the ESG weight ranges from 5.6% to 56.1%.

B. RQ2: In-Sample Validation Overstates Quality

The pipeline's former validation evidence looked strong, and none of it survives out of sample (Table III). The balanced composite's correlation with the trailing six-month return is 0.23, but those returns end on the snapshot date and overlap the price history used by the risk-adjusted and stability factors. Its correlation with the contemporaneous quality proxy is 0.50, inflated by inputs that the proxy shares with the financial, growth and operational factors. Out of sample, all three profiles have negative point ICs, and every confidence interval contains zero.

Two further designs inflated the evidence (Table IV). With the isotonic correction fitted to the quality proxy, 7 of 8 factors were "monotonic" in that proxy; without it, 3 are. The former "Bayesian bootstrap" multiplied each score by i.i.d. Exponential(1) noise and so measured the injected noise; its rank intervals had a median width of 248 positions out of 276, while the earlier draft reported perfect rank stability (Kendall $\tau = 1$) from a separate calculation. These problems affected the evidence rather than the conclusion: the pre-audit composite also has no predictive power (IC $= -0.110$).

C. RQ3: Out-of-Sample Evidence

Over the window the S&P 500 returned 17.1% and the NIFTY 50 3.2%, and the equal-weighted mid-cap universes returned 15.8% (US) and 18.9% (India). Table V reports the three primary hypotheses.

The composite does not predict returns, so H1 and H3 are not supported. Its IC with the country-neutral full-window return is -0.094 (95% CI $[-0.219, 0.036]$), and its top-19

TABLE III

IN-SAMPLE "VALIDATION" VERSUS OUT-OF-SAMPLE OUTCOME. IN-SAMPLE COLUMNS ARE RANK CORRELATIONS OF THE COMPOSITE WITH TRAILING SIX-MONTH RETURNS AND WITH A CONTEMPORANEOUS QUALITY PROXY ON THE SNAPSHOT CROSS-SECTION. OUT-OF-SAMPLE INTERVALS AS IN TABLE VI.

	In-sample (pre-snapshot)		Out-of-sample	
Composite	vs. trailing 6m	vs. quality proxy	IC, full window	95% CI
Balanced composite	0.227	0.503	−0.094	[−0.227, 0.040]
ESG-first composite	0.093	0.272	−0.111	[−0.230, 0.015]
Financial-first composite	0.288	0.580	−0.063	[−0.186, 0.073]

TABLE IV

DIAGNOSTICS BEFORE AND AFTER THE AUDIT CORRECTIONS OF SECTION III-E. PRE-AUDIT RANK INTERVALS COME FROM THE FORMER "BAYESIAN BOOTSTRAP"; POST-AUDIT INTERVALS FROM SECTION IV-E.

Diagnostic	Pre-audit	Post-audit
Factors monotonic in quintiles of the quality proxy	7/8	3/8
Median 95% rank-CI width, all firms	248	42
Maximum VIF among factor scores	2.80	2.04
Top-19 balanced portfolio: firms in common	–	13/19
Balanced composite: Spearman ρ pre vs. post	–	0.95
Out-of-sample IC of the balanced composite	−0.110	−0.094

portfolio beat the universe by 1.9 percentage points (pp), with CI $[-10.9, 11.5]$. The 9 firms acquired during the window do not change this. Assigning them the 5th or 95th percentile return puts the IC in $[-0.090, -0.083]$ and the excess return in $[-3.4, -1.3]$ pp; the excess turns negative under both bounds because the top-ranked firm (CTRA) was among the exits and the bounds keep it in the portfolio.

ESG does predict risk, so H2 is supported. Controlling for trailing volatility, higher ESG predicts lower realised volatility, with partial $\rho = -0.163$ (95% CI $[-0.276, -0.041]$, Holm $p = 0.013$). The effect is modest relative to the persistence of volatility itself: trailing volatility alone has a rank correlation of 0.82 with realised volatility, so most of the raw ESG–volatility association ($\rho = -0.30$) is past volatility carried forward.

The secondary results locate the signal (Figure 2 and Tables VI and VIII). No composite has a positive IC at any horizon, and the only Holm-significant full-window IC is negative, for the S pillar (−0.207). At one month the financial and market scores have significantly *negative* ICs, consistent with short-term reversal. In rank regressions with trailing volatility, beta, size, and sector and country effects, both the composite (-0.085, $p = 0.019$) and the S pillar (-0.103, $p = 0.014$) predict lower realised volatility, whereas the E pillar, the G pillar and the part of ESG orthogonal to the other factors do not. The G pillar's pooled association with realised beta (−0.238) disappears once sector and country effects are added (Table IX). The forward risk signal is therefore carried mainly by the financial-proxy S pillar.

The risk tilt was costly in a rising market (Table VII). The top-19 ESG-only portfolio returned −19.9 pp relative to the universe (CI $[-27.9, -10.2]$), or −17.4 pp after adjusting for trailing beta. Sector-and-country neutralisation reduces the gap

TABLE V

PRIMARY HYPOTHESES (ONE-SIDED TESTS, HOLM-ADJUSTED ACROSS H1–H3; $N = 260$ FIRMS WITH A FULL-WINDOW RETURN).

	Test	Estimate	95% CI	p	p_{Holm}
H1	IC of balanced composite with return	−0.094	[−0.219, 0.036]	0.935	1.000
H2	Partial ρ(ESG, realised vol. | trailing vol.)	−0.163	[−0.276, −0.041]	0.004	0.013
H3	Top-19 minus universe return (pp)	1.9	[−10.9, 11.5]	0.543	1.000

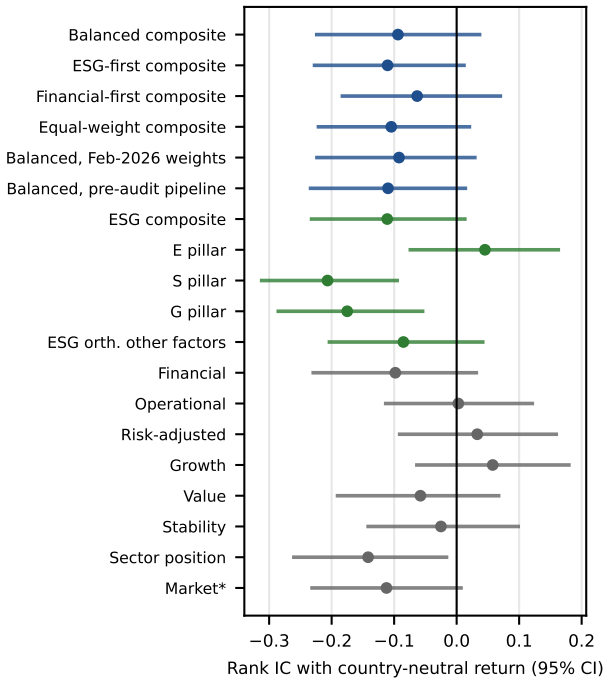


Fig. 2. Out-of-sample rank ICs over (2026-04-02, 2026-09-23] with 95% bootstrap intervals. Blue: composites; green: ESG measures; grey: financial factors (*market score carries zero weight). Two-sided 80%-power detection threshold: $|\rho| \geq 0.173$.

to −8.4 pp (CI $[-16.2, 0.5]$), so more than half of the shortfall is a sector bet: the portfolio overweights real estate, utilities and energy and holds no technology or healthcare firms, the two best-performing sectors of the window.

D. RQ4: Where the Risk Signal Comes From

Because H2 is the only supported hypothesis, we test it against the three alternative explanations of Section IV-C (Figure 3, Tables IX and X).

The signal survives known risk characteristics. With all controls and sector and country effects, the ESG coefficient is −0.106 (95% CI $[-0.190, -0.022]$, one-sided $p = 0.006$; H4). A double sort agrees: within trailing-volatility terciles, high-ESG firms had −6.1 pp lower realised volatility than low-ESG firms on average (CI $[-9.1, -3.3]$), and the gap widens from −2.8 pp in the calmest tercile to −11.7 pp in the most volatile. The result holds without the 2 takeover-pinned firms, with winsorised volatility and without utilities and real estate, and it has the same sign in both countries (US −0.208, India −0.267). The ESG–beta association does not survive full

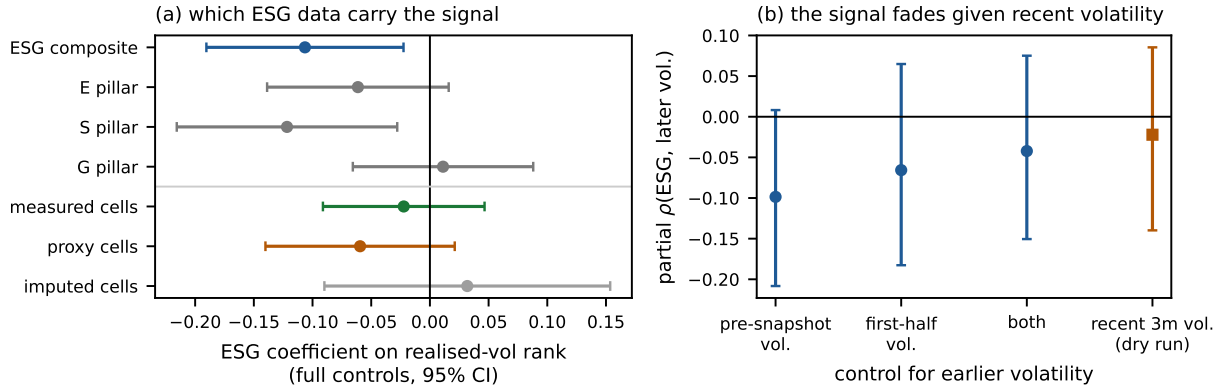


Fig. 3. Where the ESG risk signal comes from. (a) Coefficient of each ESG measure in a rank regression of realised OOS volatility with all 8 controls and sector and country fixed effects. (b) Partial Spearman correlation of ESG with realised volatility after 2026-06-28, given earlier volatility. The orange point is the registered dry run: volatility over (2026-07-01, 2026-09-23] given the preceding months. 95% intervals.

TABLE VI
OUT-OF-SAMPLE RANK ICS WITH COUNTRY-NEUTRAL RETURNS. BOLD: HOLM $p < 0.05$ WITHIN HORIZON. INTERVALS USE 1,000 RESAMPLES, SO THEY DIFFER SLIGHTLY FROM THE 5,000-RESAMPLE H1 INTERVAL IN TABLE V.

Signal	1m	3m	Full	95% CI (full)
ESG composite	−0.138	−0.078	−0.111	[−0.235, 0.016]
E pillar	−0.104	0.018	0.045	[−0.077, 0.166]
S pillar	−0.116	−0.164	**−0.207**	[−0.315, −0.092]
G pillar	−0.025	−0.118	−0.175	[−0.288, −0.052]
ESG ⊥ other factors	−0.054	−0.077	−0.085	[−0.206, 0.045]
Financial	**−0.190**	−0.099	−0.098	[−0.232, 0.034]
Operational	−0.042	0.051	0.003	[−0.116, 0.124]
Risk-adjusted	−0.038	0.096	0.033	[−0.094, 0.162]
Growth	0.032	0.038	0.058	[−0.067, 0.182]
Value	0.021	−0.120	−0.058	[−0.193, 0.070]
Stability	−0.154	−0.076	−0.025	[−0.144, 0.101]
Market (not weighted)	**−0.208**	−0.073	−0.112	[−0.234, 0.010]
Sector position	−0.098	−0.074	−0.142	[−0.263, −0.013]
Balanced composite	−0.175	−0.074	−0.094	[−0.227, 0.040]
ESG-first composite	−0.159	−0.082	−0.111	[−0.230, 0.015]
Financial-first composite	−0.168	−0.051	−0.063	[−0.186, 0.073]
Equal-weight composite	−0.160	−0.091	−0.105	[−0.224, 0.023]
Balanced, Feb-2026 weights	−0.171	−0.078	−0.092	[−0.226, 0.032]
Balanced, pre-audit pipeline	**−0.189**	−0.091	−0.110	[−0.237, 0.017]

TABLE VII
TOP-19 PORTFOLIO EXCESS RETURN OVER THE EQUAL-WEIGHTED UNIVERSE (PP, COUNTRY-NEUTRAL). β-ADJ.: MARKET-MODEL ADJUSTMENT WITH TRAILING BETA; SECT.-NEUT.: SECTOR-AND-COUNTRY-NEUTRAL FULL-WINDOW RETURNS.

Top-N by	1m	3m	Full	β-adj.	Sect.-neut.	95% CI (full)
Balanced composite	2.2	0.9	1.9	1.6	3.1	[−11.2, 11.0]
ESG-first composite	−0.6	−5.7	−9.4	−8.6	−0.8	[−18.5, 0.8]
Financial-first composite	1.5	2.5	0.9	0.3	−1.3	[−10.5, 13.8]
Equal-weight composite	0.3	−0.9	−2.8	−3.8	1.0	[−11.4, 7.7]
Balanced, Feb-2026 weights	0.5	−1.0	2.1	1.3	6.4	[−9.7, 16.2]
Balanced, pre-audit pipeline	1.2	−2.3	−3.4	−3.2	−2.8	[−12.6, 4.9]
ESG composite	−1.1	−8.0	−19.9	−17.4	−8.4	[−27.9, −10.2]

controls (-0.033, $p = 0.562$), and no ESG measure predicts drawdowns.

The signal does not come from measured ESG data. It

TABLE VIII
ESG AND REALISED RISK. PARTIAL ρ CONTROLS FOR THE TRAILING VALUE OF THE SAME RISK MEASURE; BOLD: HOLM $p < 0.05$ ACROSS THE 15 SECONDARY TESTS. REGRESSION: RANK OF REALISED VOLATILITY ON ESG RANK WITH TRAILING VOLATILITY, BETA, SIZE, AND SECTOR AND COUNTRY EFFECTS (HC3).

	Realised volatility		Realised β	Vol. regression	
ESG measure	raw ρ	partial ρ	partial ρ	coef.	p
ESG composite	−0.303	−0.163	**−0.239**	−0.085	0.019
E pillar	−0.151	−0.071	−0.067	−0.052	0.142
S pillar	−0.285	−0.123	**−0.330**	−0.103	0.014
G pillar	−0.252	−0.125	**−0.238**	0.005	0.908
ESG ⊥ other factors	−0.184	−0.154	−0.071	−0.056	0.093

sits in the S pillar (-0.122, $p = 0.011$), whose inputs are almost all financial proxies, while the G pillar carries none of it (0.011). Splitting each firm's cells by source makes the point directly. Measured cells (ISS, SEC, registry) are nearly independent of the financial factors ($R^2 = 0.07$) and have partial $\rho = 0.032$ with realised volatility (CI $[-0.10, 0.16]$). Proxy cells are largely financial ($R^2 = 0.57$) and have the same sign as the S pillar, though imprecisely ($\rho = -0.086$, CI $[-0.20, 0.04]$). Imputed cells show a strong raw association (-0.269) that vanishes once sector fixed effects are included. Adding measured-cell ESG to the out-of-fold volatility forecast changes R^2 by -0.0021.

The signal is small and partly stale volatility. Adding the composite to a cross-validated volatility forecast raises out-of-fold R^2 from 0.729 to 0.734, an improvement in 100% of repeats but only 0.0056 in size. Given pre-snapshot volatility, ESG's partial correlation with second-half volatility is -0.099; given first-half realised volatility, which an investor would have observed, it falls to -0.066 (CI $[-0.18, 0.06]$, $p = 0.292$). The monthly partial correlations are negative in 6 of 6 months but small. ESG therefore behaves like a slow-moving summary of a firm's risk profile, adding a little

TABLE IX

RANK-REGRESSION COEFFICIENTS OF EACH ESG MEASURE FOR REALISED RISK. "TRAILING": CONTROL FOR THE PRE-SNAPSHOT VALUE OF THE SAME RISK MEASURE; "FULL": ALL CONTROLS OF SECTION IV-C, WITH SECTOR AND COUNTRY FIXED EFFECTS (HC3). DRAWDOWN IS SIGNED, SO A POSITIVE VALUE MEANS A SHALLOWER DRAWDOWN. BOLD: $p < 0.05$ (TWO-SIDED, UNADJUSTED). $N = 260$.

	Realised vol.		Realised β		Drawdown
ESG measure	trailing	full	trailing	full	full
ESG composite	**−0.087**	**−0.106**	**−0.138**	−0.033	0.060
E pillar	−0.048	−0.061	**−0.103**	0.000	0.063
S pillar	**−0.103**	**−0.122**	**−0.114**	0.035	0.089
G pillar	−0.001	0.011	−0.055	−0.043	−0.088
Measured-cell ESG	−0.018	−0.022	−0.064	−0.021	−0.069
Proxy-cell ESG	−0.067	−0.059	**−0.144**	0.048	0.071
Imputed-cell ESG	0.025	0.032	0.047	−0.056	0.094

TABLE X

ROBUSTNESS OF THE ESG–VOLATILITY RESULT. UPPER PANEL: H2 AND H4 ON SUBSAMPLES. LOWER PANEL: PARTIAL SPEARMAN CORRELATION OF ESG WITH REALISED VOLATILITY AFTER 2026-06-28, GIVEN EARLIER VOLATILITY.

Sample / test	N	H2 partial ρ [95% CI]	H4 coef. [95% CI]
all (pre-declared)	260	−0.163 [−0.27, −0.04]	−0.106 [−0.19, −0.02]
excl. deal-pinned	258	−0.166 [−0.27, −0.04]	−0.104 [−0.18, −0.03]
winsorised vol 1/99	260	−0.163 [−0.27, −0.05]	−0.106 [−0.19, −0.02]
US only	172	−0.208 [−0.35, −0.06]	−0.067 [−0.17, 0.04]
India only	88	−0.267 [−0.46, −0.04]	−0.174 [−0.41, 0.06]
excl. Utilities, Real Estate	233	−0.171 [−0.29, −0.04]	−0.110 [−0.20, −0.02]
2nd-half vol. | pre-snapshot vol.	260	−0.099 [−0.21, 0.01]	–
2nd-half vol. | 1st-half vol.	260	−0.066 [−0.18, 0.06]	–
2nd-half vol. | both	260	−0.042 [−0.15, 0.08]	–

beyond long-horizon trailing volatility and little beyond recent volatility.

E. Robustness of the Return Null

The failure of H1 does not depend on the chosen weights (Figure 4). Across 2000 weight vectors drawn uniformly from the simplex, the OOS IC has median −0.087 and is positive in only 3.3% of draws. More ESG weight lowers the IC, which moves from −0.055 with no ESG to −0.111 with ESG only. The low risk of the index portfolio does not come from ESG either: its realised volatility is below the universe's in 98% of draws whatever the ESG weight, and already −7.0 pp relative to the universe with no ESG weight, because the risk-adjusted and stability factors select calm firms. Monthly ICs of the balanced composite are negative in 3 of 6 months, and round-trip costs of 0.21 pp leave a net excess of 1.7 pp (1.3 pp at three times the costs), which does not change the H3 conclusion.

F. Rank Uncertainty

Given the construction, rankings near the top are reasonably well determined (Figure 5 and Table XI). For the top-19 names, the median 95% rank interval spans 15 ranks, and weight uncertainty dominates sampling uncertainty (13 against 3 ranks). Of the 19 current constituents, 18 remain in the top-19 in at least half of the draws, and a perturbed portfolio is expected to share 16.5 names with the published one. Stable rankings do not, however, make the construction predictive.

TABLE XI

TOP-19 BALANCED PORTFOLIO: 95% RANK INTERVAL, PROBABILITY OF REMAINING IN THE TOP-19, AND REALISED FULL-WINDOW RETURN (†NSE; "EXIT": ACQUIRED DURING THE WINDOW).

Rank	Ticker	Sector	95% CI	P(top-N)	OOS ret. (%)
1	CTRA	Energy	[1, 2]	1.00	exit
2	INCY	Healthcare	[1, 2]	1.00	29.9
3	GGG	Industrials	[3, 6]	1.00	−6.7
4	DECK	Consumer Cyclical	[3, 11]	1.00	−20.1
5	STLD	Basic Materials	[3, 10]	1.00	30.3
6	NAVINFLUOR†	Basic Materials	[3, 8]	1.00	43.7
7	STAG	Real Estate	[4, 8]	1.00	1.8
8	NNN	Real Estate	[6, 27]	0.93	−0.7
9	EWBC	Financial Services	[6, 21]	0.96	15.7
10	PIDILITIND†	Basic Materials	[6, 16]	1.00	25.6
11	MUTHOOTFIN†	Financial Services	[7, 19]	0.99	−9.4
12	MTCH	Communication Services	[9, 30]	0.68	30.3
13	CDNS	Technology	[9, 25]	0.87	10.9
14	IBKR	Financial Services	[8, 31]	0.76	35.4
15	ITT	Industrials	[10, 25]	0.84	10.1
16	VEEV	Healthcare	[9, 31]	0.73	55.6
17	RPM	Basic Materials	[11, 29]	0.52	3.2
18	INVH	Real Estate	[11, 39]	0.42	6.5
19	ATO	Utilities	[9, 34]	0.60	−16.4

VI. DISCUSSION

A. Public-Data ESG Is Mostly Financial Quality

The construct-validity result has a simple explanation. When direct disclosure is missing, each proxy is chosen for a plausible link between a financial ratio and an ESG outcome, such as margins and emissions [6], revenue per employee and satisfaction [7], or cash returned and community investment. Each link is defensible in isolation, but together they turn the E and S pillars into a within-sector profitability and payout score. That score is largely spanned by the financial factors, and it is what carries the index's forward risk signal. The G pillar, built on third-party governance assessments, is the only component that adds roughly orthogonal information, although that information did not predict risk or returns in our window. Index builders should therefore report provenance and redundancy per pillar alongside coverage, and should not describe proxy E and S pillars as environmental or social performance.

B. Why In-Sample Evidence Misled

The four validation pitfalls of Section V-B share one feature: the validation target was not independent of what was being validated. Trailing returns overlap the price history used by the risk-adjusted and stability factors; the "forward" quality proxy shares inputs with the factors, and the isotonic correction was fitted to it; and the bootstrap perturbed scores rather than data, so it measured its own noise. Each of these designs yields favourable numbers whether or not the index works, which is the mechanism behind many replication failures in factor research [14], [15]. The defence is procedural: freeze the construction at a date and test it on outcomes that occur after that date.

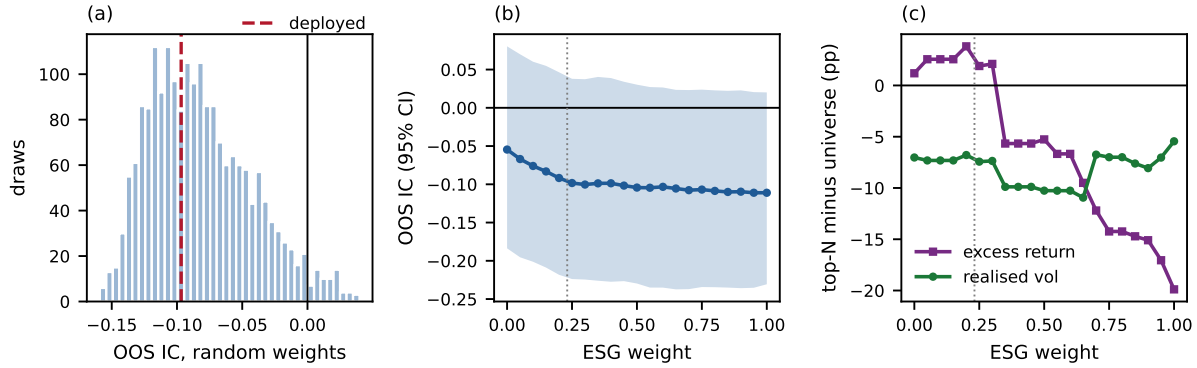


Fig. 4. Weight uncertainty and the ESG weight. (a) OOS IC of the composite under 2000 weight vectors drawn uniformly from the simplex; dashed line: deployed weights. (b) OOS IC as the ESG weight rises from 0 to 1, with the other weights in their deployed proportions (95% bootstrap band). (c) The top-19 portfolio's excess return and realised volatility relative to the universe on the same sweep. Dotted line: deployed ESG weight.

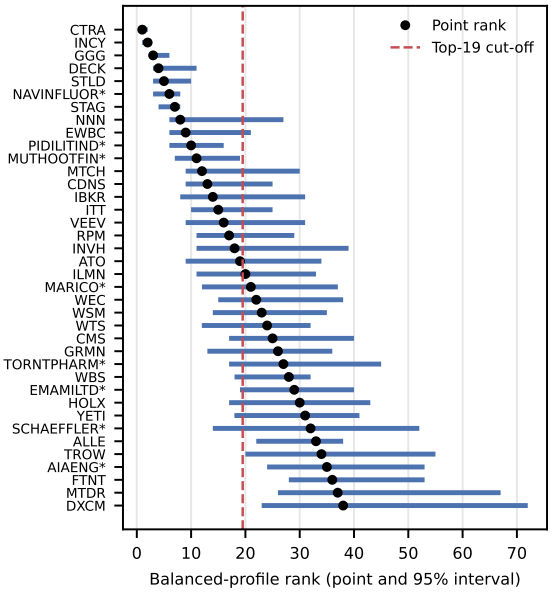


Fig. 5. Balanced-profile ranks with 95% intervals combining universe resampling and Dirichlet ($\kappa = 100$) weight perturbation ($B = 1000$). *NSE-listed.

C. ESG as a Risk Tilt, and Its Price

The forward evidence supports a risk channel, but a narrow one. Higher ESG predicts lower volatility beyond 8 risk characteristics, including trailing volatility, in line with [12], but it does not predict beta or drawdowns once sector and country are controlled. The decomposition of Section V-D also limits what the channel can be credited to, because the signal comes from the proxy-heavy S pillar rather than from measured cells and is largely absorbed by recent realised volatility. The defensible reading is that public-data ESG, as built here, is a slow-moving low-risk characteristic assembled from financial data, not evidence that environmental or social performance lowers risk.

The evidence also does not support a return channel over six months. In a window in which both markets rose, the risk tilt was expensive, because an ESG-only screen loaded on real estate, utilities and energy and held no technology or healthcare firms. This is the trade-off that equilibrium models predict, in which sustainable assets serve as hedges while offering lower expected returns [20], [21]. A single six-month window cannot estimate a premium, but it does show the size of the regime risk that an ESG-tilted mid-cap portfolio takes on relative to its universe.

D. Implications for Index Builders

Three practices follow from the evidence and cost little to adopt. First, provenance per cell and redundancy per pillar should be standard disclosures for any ESG index built from public data. Second, the construction should be frozen and dated, with any number computed on the snapshot cross-section treated as descriptive, and out-of-sample results should be reported with their minimum detectable effect, since a few hundred firms can reject only large effects. Third, before an out-of-sample effect is credited to ESG, it should be re-estimated separately on measured, proxy and imputed cells and controlled for recent risk. Generating the paper from the code also helps: every discrepancy between manuscript and pipeline that our audit found was a number copied or written by hand.

VII. Limitations and Threats to Validity

The main limitation is that the OOS test covers one window of 118 trading days in a rising market. With $N = 260$, only $|\rho| \geq 0.173$ is detectable at 80% power, so H1 and H3 exclude large effects rather than small ones, and the risk result may not generalise to other regimes. In addition, the hypotheses and analysis code were fixed before OOS prices were downloaded but after the calendar window had passed, the profile weights were revised inside the window, and the authors' general knowledge of market conditions cannot be excluded. The frozen snapshot artefacts and the full reporting of alternative

weightings limit these risks, since the February 2026 weights and 96.7% of random weightings give the same return conclusion. The mechanism analyses of Section V-D are exploratory. The second window (Section IV-D) was registered before it opened precisely so that H2 and H4 can be confirmed or refuted there.

The findings concern public-data ESG as constructed here. Our E and S pillars are not measurements of environmental or social performance, and the results need not transfer to vendor ratings, which have their own validity problems [2], [5]. The measured cells are mostly governance items, so the null for measured cells is not a test of measured environmental data. The governance pillar itself depends on ISS scores redistributed by a free data interface, which cover 209 of 269 firms and may not remain available. The provenance subscores also average indicator ranks with equal weights, so they show where the signal lives rather than how the composite would behave with a single data source.

Three design choices affect generality. The candidate list was taken from index constituents at the time of collection rather than from a point-in-time membership history, so survivorship before the evaluation window cannot be excluded. US and Indian firms are scored against a common reference that includes 45 large caps; country-neutral returns remove market-level differences in outcomes but not in disclosure regimes, and with $N = 88$ the Indian estimates are imprecise, although they have the same sign as the US ones (Table X). Finally, portfolio results are equal-weighted and single-period, with stylised round-trip costs and no model of market impact at scale.

VIII. Conclusion

We built a transparent nine-factor ESG–financial index for 269 US and Indian mid-caps and audited it end to end. Its E and S pillars turn out to be mostly within-sector financial quality, with other factors explaining 0.42 and 0.51 of their variance, and only the governance pillar carries largely independent information. The in-sample evidence that made the index look strong came from validation targets that were not independent of the index. Out of sample, the composite does not predict returns (IC $= -0.094$), but higher ESG predicts lower future volatility beyond trailing volatility (partial $\rho = -0.163$, Holm $p = 0.013$) and beyond 8 known risk characteristics and sector effects. That risk signal comes from the proxy-built S pillar rather than from measured cells (0.032), adds 0.0056 to an out-of-fold volatility forecast, and falls to -0.066 given recent realised volatility. In a rising market the tilt was costly, with an ESG-only top-19 screen returning -19.9 pp relative to the universe.

The contribution is as much methodological as empirical. Provenance per cell, redundancy per pillar, a dated construction and an out-of-sample test with fixed hypotheses are cheap to implement, and together they separate a transparent composite from a valid one. The natural next step is the pre-registered window (2026-09-28 to 2027-03-31), which will confirm or refute the risk result against recent-volatility controls. Beyond it, the same snapshot–hold-out cycle can be repeated each quarter to accumulate power, and replacing the financial proxies with measured E and S data, such as EPA TRI records, the Business Responsibility and Sustainability Reports (BRSR) of Indian firms and text-mined sustainability reports [29], would show whether genuine environmental and social measurement shares the risk signal.

Reproducibility. The code, configuration, cached raw data, frozen pre-audit outputs and the registration package are in the project repository (https://github.com/Shashwat1729/esg-hybrid-index). A single command rebuilds every table, figure and number in this paper from the 2026-04-02 snapshot and the cached post-snapshot prices, and the repository documents each step, including the evaluation of the registered window once it has closed.

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APPENDIX A
ESG INDICATORS AS ACTUALLY CONSTRUCTED

Table XII lists every ESG indicator that carries cross-sectional variation in the mid-cap universe, together with the source that actually fills it. "Sector pct." means the within-sector percentile of the stated ratio, clipped to $[2, 98]$. When a source is missing, a cell falls through to the next tier of Section III-B. The other 13 declared indicators have no source for any mid-cap firm and are excluded from scoring. They are: Scope 2 and 3 emissions, emissions intensity, energy efficiency, water intensity, environmental fines, R&D intensity (E innovation), employee turnover, women in management, pay gap, injury rate, CEO pay ratio and the anti-corruption policy flag.

APPENDIX B
SASB MATERIALITY WEIGHTS

Table XIII gives the pillar weights used to combine the E, S and G scores within each sector (Section III-C). Each row sums to one.

APPENDIX C
AUDIT LOG

All corrections below were made before any post-snapshot price was downloaded.

1) *ESG direction.* The Scope 1 emissions and ESG risk-rating indicators were removed from the lower-is-better set, because both are stored as higher-is-better scores.
2) *Imputation noise.* Multiplicative $\mathcal{N}(1, 0.04)$ noise in sector and global median imputation was removed.
3) *Registry data.* Registry memberships are encoded as 0/1 and no longer overwrite continuous emissions and renewable-share indicators. A mapping of a 40–70 score onto the binary anti-corruption flag, which made that flag constant, was removed.
4) *Source precedence.* ISS governance scores now take precedence over SEC financial items mapped onto governance indicators. A coding error in this step had prevented the data-collection stage from running.
5) *Monotonicity correction.* The isotonic fit to the validation proxy is disabled in all reported results.
6) *Preference transforms.* The undocumented bottom-30% quality gate and the power exponent of 1.5 are disabled in all reported results.

TABLE XII
SOURCE OF EACH INFORMATIVE ESG INDICATOR (MID-CAP UNIVERSE).

Pillar	Indicator	Actual construction
E	Scope 1 emissions	Sector pct. of operating margin (higher margin ⇒ assumed lower emissions intensity) [6]
E	Renewable-energy share	Sector pct. of revenue / market capitalisation
E	Waste recycling rate	Sector pct. of gross margin [27]
E	Carbon-reduction target	1 if listed as CDP reporter, SBTi participant or RE100 member, else 0
S	Gender diversity	Sector pct. of R&D / revenue (operating-margin fallback)
S	Safety training hours	Mean of sector pct. of log employees and of operating margin
S	Employee satisfaction	Sector pct. of revenue per employee [7]
S	Revenue per employee	Direct ratio; duplicates the previous row's input
S	Community investment	Sector pct. of dividend yield + FCF / revenue
S	Supply-chain audit coverage	Sector pct. of payout ratio + current ratio [46]
S	Human-rights policy	1 if UN Global Compact signatory, else 0
G	Board independence	ISS board risk, $(10 - r) \times 10$; else SEC dividend item; else sector median
G	ESG-linked executive pay	ISS compensation risk; else SEC share-based compensation item
G	Shareholder rights	ISS shareholder-rights risk; else sector pct. of $1/(\text{D/E})$ + dividend yield
G	Ethics and compliance	ISS audit risk; else sector median
G	ESG risk rating	ISS overall governance risk; else sector median
G	Tax transparency	SEC effective tax rate (banded); else institutional ownership (%)
G	Data privacy	Insider ownership, banded (peak score at 10–30%)
G	ESG controversies	Analyst coverage, $20 + 2.5\times$ count, capped at 90
G	Board diversity	Sector pct. of log market cap and inverse beta [47]
G	Board size	SEC operating-segment count, banded (8 firms); else sector or global median

TABLE XIII
SECTOR-SPECIFIC E/S/G PILLAR WEIGHTS, INFORMED BY THE SASB MATERIALITY MAP [10].

Sector	E	S	G
Technology	0.20	0.45	0.35
Healthcare	0.25	0.40	0.35
Financial Services	0.15	0.35	0.50
Energy	0.55	0.20	0.25
Industrials	0.40	0.30	0.30
Consumer Cyclical	0.25	0.40	0.35
Consumer Defensive	0.30	0.35	0.35
Basic Materials	0.50	0.25	0.25
Utilities	0.50	0.20	0.30
Real Estate	0.40	0.25	0.35
Communication Services	0.20	0.45	0.35

7) *Snapshot consistency.* FX is taken at the snapshot rate. Within-sector revenue share is recomputed after currency conversion. Firms that did not trade on the snapshot date are removed.
8) *Portfolio size.* $N^\star = \lceil 0.07 N_{\text{mid}} \rceil$, bounded to $[10, 50]$, replaces an earlier rounding rule that hard-coded a special case for a 276-firm universe.
9) *Rank uncertainty.* The score-perturbing "Bayesian bootstrap" was replaced by the procedure of Section IV-E.
10) *Manuscript.* Several results in an earlier draft could not be reproduced from the code and were removed. These were walk-forward ICs, Fama–MacBeth estimates, an ablation table, a claimed sign flip of the value factor, and "100% of periods" out-performance of a momentum proxy. Two further claims were withdrawn:
    - a large-cap "generalisation" statistic ($\rho \approx 1$), which measured only the effect of adding 45 firms to the percentile reference set;
    - a Kendall's W that measured agreement between investor profiles, not generalisation.

A residualisation of E on the financial score, described in the draft, was never executed by the pipeline and is not claimed here.